

Advanced Mathematical Tools for Management

MATH 40604 — Study Notes

Khaled Fouda

6 August 2026

Table of contents

How to use these notes	3
The structure of each chapter	3
Notation used throughout	3
A note on effort	4
1 Modelling, Sets, and Algebra	5
This week at a glance	5
2 What a mathematical model is	6
2.1 The components of an optimization problem (decision model)	6
2.2 Deterministic and probabilistic models	8
2.3 Statistical models	9
3 Sets and subsets	11
3.1 Basic mathematical notation	11
3.2 Describing a set	12
3.3 The standard number sets	13
3.4 Absolute value of a number	14
3.5 Summation notation (sigma)	15
3.6 Subsets, cardinality, and set operations	20
4 Essential algebra for working with models	24
4.1 Algebraic expressions	24
4.2 Polynomials and quadratic equations	24
4.3 Exponents and logarithms	26
5 Proof techniques, propositions, and quantifiers	29
One-minute recap	30

How to use these notes

These notes accompany **MATH 40604, Advanced Mathematical Tools for Management**. They are a study aid, not a replacement for the lectures: the official course material is the slide deck distributed each week. What the slides compress — the intermediate algebra, the reason a definition is written the way it is, the mistake that costs marks — is what these notes expand.

They are written on one assumption: that you are a capable adult who has not done mathematics in a while. Nothing here requires you to remember a formula from years ago. Where a step looks small, it is written out anyway.

The structure of each chapter

Every chapter follows the same path.

- **Definitions** state an idea precisely, and are always followed by an *In words* reading in plain English. If the symbols do not land, read the plain-English version first and come back.
- **Key results** are the facts you will actually use. Each one is derived, not asserted.
- **Worked examples** are set in a business or analytics context and show every algebraic step.
- **Common pitfalls** collect the errors that recur — usually not from misunderstanding the concept, but from a small slip in notation.
- **One-minute recap** is what to reread on the train.

Notation used throughout

Mathematical notation varies between textbooks, countries, and software. The conventions below are fixed for the whole course; where a different convention is common elsewhere, it is noted.

Symbol	Meaning	Note
$\mathbb{N}$	natural numbers $\{0, 1, 2, \dots\}$	includes zero in this course
$\mathbb{Z}$	integers $\{\dots, -1, 0, 1, \dots\}$	
$\mathbb{Q}$	rational numbers	quotients p/q with $q \neq 0$
$\mathbb{R}$	real numbers	
$\mathbb{R}_+$	non-negative reals, $[0, +\infty)$	includes 0
$\mathbb{R}^p$	real vectors with p components	
$\mathbb{R}^{n \times p}$	real matrices, n rows and p columns	
(a, b)	open interval	parentheses , not $]a, b[$
$[a, b]$	closed interval	

Symbol	Meaning	Note
Δ_p	the simplex in $\mathbb{R}^p$	p components, non-negative, summing to 1
$\mathbf{x}$	a vector	bold , always
n	number of observations	never anything else
p	number of variables	never anything else
$\bar{x}$	mean of an observed sample	
μ	mean of a population or distribution	not the same object as $\bar{x}$
$\ln$	natural logarithm, base e	written $\ln$, never $\log$
$\log_{10}$	logarithm base ten	always written with its base

Four of these are worth pausing on, because they are the ones that silently change answers.

Zero is a natural number here. Many textbooks, particularly North American ones, start $\mathbb{N}$ at 1. This course starts at 0. So “ $0 \in \mathbb{N}$ ” is *true* in this course and false in some books you may own. When in doubt, look at the convention table rather than at your memory.

Open intervals use parentheses. French-language material writes the open interval from a to b as $]a, b[$. This course writes (a, b) , the convention used in statistics and data science software. You will see the French form on some slides; it means the same thing.

“Non-negative” and “positive” are different words. *Positive* means strictly greater than zero. *Non-negative* means greater than or equal to zero. French uses *positif* for both, so translated material sometimes says “positive” where “non-negative” is meant. When it matters, this course says which.

$\ln$ and $\log$ are not interchangeable. In R, $\log()$ is the *natural* logarithm and $\log_{10}()$ is base ten. In many spreadsheets and in engineering texts, **LOG** means base ten. These notes always write $\ln$ for the natural logarithm and always write the base explicitly otherwise.

A note on effort

You will get more out of a worked example by covering the solution and attempting it than by reading it. The examples are chosen so that this is realistic: none of them requires an insight you have not been given.

1 Modelling, Sets, and Algebra

This week at a glance

By the end of this chapter you should be able to:

- take a described business situation and say which quantities are **variables** and which are **parameters**, and justify the split;
- write down the objective and constraints of a small decision problem in symbols;
- describe a set three ways — by listing, by a defining property, and as an interval — and move between them;
- read and write $\cup$, $\cap$, complement, and difference, and compute the size of a union without double-counting;
- expand and condense $\sum$ notation, and prove the two identities about deviations from the mean;
- solve linear and quadratic equations, and use logarithms to turn a multiplicative relationship into a linear one.

Nothing here is new mathematics. Most of it you have met before and put down. The aim is to pick it up in a form you can use.

2 What a mathematical model is

A **mathematical model** is a simplified description of a real-world situation, written in symbols. We say simplified because a mathematical model leaves things out on purpose. For example, a model of a delivery network that accounted for every pothole would be useless, because you could not solve it and could not learn anything from it.

The mathematical model is represented by an equation between variables and parameters. A **parameter** is a quantity whose value is settled before you solve the model. It is either an input given in the problem statement, measured, or estimated from the data. A **variable** is a quantity whose value is produced by solving the model, or supplied by an observation.

Consider a straight-line relationship $y = \beta_0 + \beta_1 x$. If you are a manager who has been handed $\beta_0 = 12$ and $\beta_1 = 3$ and asked to predict y when $x = 20$: then β_0, β_1 are **parameters** and y, x are **variables**. If instead you are an analyst holding a spreadsheet of (x, y) pairs and trying to work out what β_0 and β_1 are, the roles swap: the data are now the inputs and the two coefficients are what you are solving for. The same symbol can play either role. Which one it plays depends on the question being asked, so keep the question in view and the classification is never ambiguous.

2.1 The components of an optimization problem (decision model)

Example

A company manufactures two products, A and B . It must determine the quantities to produce each week in order to maximize its profit.

Product A generates a profit of \$30 per unit and requires 2 hours of labor. Product B generates a profit of \$20 per unit and requires 1 hour of labor. The company has 100 labor hours available per week. Because of demand limitations, it cannot sell more than 40 units of A or more than 70 units of B .

Setting it up. What can the company choose? Only how much of each product to make. The profits, the labor times, the weekly capacity and the two demand ceilings are all handed to it. So the decisions are

$$x_A = \text{units of } A \text{ produced per week}, \quad x_B = \text{units of } B \text{ produced per week}.$$

Total weekly profit is \$30 for every unit of A plus \$20 for every unit of B :

$$\text{maximize } 30x_A + 20x_B.$$

Labor is the binding resource. Each unit of A consumes 2 hours and each unit of B consumes 1 hour, and only 100 hours exist:

$$2x_A + x_B \leq 100.$$

Demand limits what can be sold, and negative production is meaningless:

$$x_A \leq 40, \quad x_B \leq 70, \quad x_A \geq 0, \quad x_B \geq 0.$$

Solving it. Labor is the scarce resource, so compare the two products on profit *per labor hour* rather than per unit:

$$A : \frac{\$30}{2 \text{ h}} = \$15 \text{ per hour}, \quad B : \frac{\$20}{1 \text{ h}} = \$20 \text{ per hour}.$$

This reverses the impression given by the unit profits. Product A earns more per unit, but B earns more per hour of the resource that is actually scarce — so make as much B as can be sold:

$$x_B = 70, \quad \text{consuming } 70 \times 1 = 70 \text{ hours}.$$

That leaves $100 - 70 = 30$ hours for A , and each unit of A needs 2 hours:

$$x_A = \frac{30}{2} = 15.$$

The resulting profit is

$$30(15) + 20(70) = 450 + 1400 = \$1,850.$$

Check the answer is legal. Labor: $2(15) + 70 = 100 \leq 100$, satisfied exactly — the constraint is *binding*. Demand for A : $15 \leq 40$, satisfied with room to spare. Demand for B : $70 \leq 70$, binding. Both variables non-negative. All constraints hold, so the plan is feasible.

Is it the best? Compare it with the alternative of favouring A . Producing $x_A = 40$ leaves $100 - 80 = 20$ hours, hence $x_B = 20$, for a profit of $30(40) + 20(20) = \$1,600$ — worse by \$250. The profit-per-hour comparison told us this in advance.

Now the anatomy, using that example:

Component	What it is	In the example
Decision variables	The quantities you choose. Unknown before solving.	x_A, x_B
Parameters	Values handed to you. Known before solving.	unit profits 30 and 20; labor times 2 h and 1 h; capacity 100 h; demand ceilings 40 and 70
Objective function	The single number you are making as large or as small as possible.	$30x_A + 20x_B$, maximized
Constraints	Equations and inequalities every admissible answer must satisfy.	$2x_A + x_B \leq 100$; $x_A \leq 40$; $x_B \leq 70$; $x_A, x_B \geq 0$

Component	What it is	In the example
Assumptions	Simplifications that make the problem solvable, and that you should be prepared to defend.	unit profits constant; labor times known; maximum demand known; every unit produced can be sold

That last row deserves a comment, because assumptions are the part of a model that does its work silently. “Every unit produced can be sold” is what lets us treat the demand ceilings as the only sales limit; drop it and unsold inventory becomes a cost the model never sees. And nothing in the mathematics stops x_A from coming out as 15.5 units. Here the answer happened to land on whole numbers. In general, forcing variables to be whole makes a problem substantially harder — it becomes an *integer* program. Recognising when you have made that assumption matters more right now than knowing how to fix it.

Common pitfall

Confusing the objective with a constraint. “Use no more than 100 labor hours” and “make as much profit as possible” do different jobs. The first is a bar you must not cross — a constraint, written with $\leq$ and a fixed number on the right. The second is what you optimise, and it has no number on the right at all.

A quick diagnostic: a constraint has a comparison symbol ($\leq$, $\geq$, $=$) and a fixed number. An objective has the word *maximize* or *minimize* and no comparison. If you have written “maximize profit ≤ 1850 ”, you have merged the two.

Common pitfall

Ranking options by the wrong unit. Product *A* is the more profitable product *per unit* and the less profitable one *per labor hour*. When a resource is scarce, it is profit per unit of that resource that decides. Choosing *A* because $\$30 > \20 costs \$250 a week here.

2.2 Deterministic and probabilistic models

The production model is **deterministic**: profits, production times, capacity and maximum demand are all treated as known with certainty. Put the same inputs in twice and you get the same answer twice.

That is a strong assumption, and often a false one. The demand ceiling of 40 units for product *A* is really a forecast. A **probabilistic** (or *stochastic*) model replaces the fixed value with a random quantity:

$$D_A \sim \mathcal{N}(40, 5^2),$$

read as “demand for *A* follows a normal distribution with mean 40 and standard deviation 5”.

The company would then have to choose its production level while accounting for the risk of producing more than the realized demand — a cost the deterministic model cannot represent, because in it demand is never disappointing.

⚠ Common pitfall

In the notation $\mathcal{N}(\mu, \sigma^2)$ the **second argument is the variance, not the standard deviation**. Writing $\mathcal{N}(40, 5^2)$ means a standard deviation of 5. Writing $\mathcal{N}(40, 5)$ means a *variance* of 5, hence a standard deviation of about 2.24 — a different model. Some software, including R's `rnorm`, takes the standard deviation as its argument instead, which is a frequent source of error when moving between a formula and code.

2.3 Statistical models

A third kind of model does not choose anything. It explains: given data, how does one quantity move with another?

Example

Suppose we want to explain a company's sales as a function of its advertising expenditures. For period i we record sales Y_i and advertising spend X_i , and propose

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

Reading the pieces:

- Y_i is the company's sales in period i — the quantity we are trying to account for.
- X_i is advertising expenditure in period i .
- β_0 is baseline sales: what the model says would be sold with no advertising at all.
- β_1 is the effect of advertising — the change in sales associated with one extra dollar spent.
- ε_i is everything the model does not capture: a competitor's promotion, the weather, a supply problem. It is not an error in the sense of a mistake; it is the part of Y_i that advertising does not explain.
- σ^2 is the error variance, measuring how much that unexplained part varies.

This is where the vocabulary of *dependent* and *independent* variables belongs, and it is worth defining properly rather than mentioning in passing.

i Definition: response and explanatory variables

In a statistical model, the **response variable** (also called the **dependent variable**) is the quantity the model sets out to explain or predict. It sits on the left-hand side.

An **explanatory variable** (also called an **independent variable**, a *predictor*, a *regressor*, or a *covariate*) is a quantity used to account for variation in the response. Explanatory variables sit on the right-hand side.

In words: the response is what you want to know; the explanatory variables are what you already know and are using to guess it.

In the sales model, Y_i is the response and X_i is the explanatory variable. The names “dependent” and “independent” are older and somewhat misleading — “independent” here has nothing to do with statistical independence, and explanatory variables are frequently related to one another. This course prefers *response* and *explanatory*, but you will meet all of these words and they mean the same thing.

 Common pitfall

Do not use “response” and “explanatory” for a model with no random component.

They are statistical terms; they presuppose data and unexplained variation. In a deterministic model — a supply-and-demand equilibrium, say, where you solve algebraically for a price — the right distinction is between quantities determined *inside* the model (**endogenous**) and quantities taken as given from *outside* it (**exogenous**). Calling a price an “explanatory variable” when the whole point is to solve for it is a contradiction: it cannot both explain and be the unknown.

One notational point, which now has a reason attached. The quantities $\beta_0, \beta_1, \sigma^2$ are properties of the population and are unknown; the numbers you compute from a sample are *estimates* of them, written $\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2$. The same distinction separates μ , a population mean, from $\bar{x}$, a sample mean. They are different objects and the notation should not blur them.

3 Sets and subsets

A **set** is a well-defined collection of objects, called its **elements**. Sets are useful in management and analytics because so many things are naturally collections: a population, a customer segment, the domain of a variable, the set of feasible decisions, a set of observations.

We write $x \in A$ for “ x belongs to the set A ”, and $x \notin A$ for “ x does not belong to A ”.

3.1 Basic mathematical notation

Symbol	Reading	Example
$a = b$	a is equal to b	$x = 3$
$a \neq b$	a is not equal to b	$x \neq 0$
$a < b$	a is less than b	$x < 10$
$a \leq b$	a is less than or equal to b	$x \leq 10$
$a > b$	a is greater than b	$x > 10$
$a \geq b$	a is greater than or equal to b	$x \geq 10$
$a \in A$	a belongs to the set A	$\sqrt{2} \in \mathbb{R}$
$a \notin A$	a does not belong to A	$-1 \notin \mathbb{N}$
$\forall x$	for every x , for all x , for any x	$\forall x \in \mathbb{R}$
$\exists$	there exists	$\exists x > 0$
$\Rightarrow$	implies	$P \Rightarrow Q$
$\Leftrightarrow$	is equivalent to, if and only if	$P \Leftrightarrow Q$

The last two are the ones to slow down on, because they are not symmetric with each other.

Example

An implication. Let total revenue be

$$R = pq,$$

where p is unit price and q is quantity sold. If the selling price rises while demand is unchanged, revenue rises:

$$(p_1 > p_0 \text{ and } q_1 = q_0) \Rightarrow R_1 > R_0.$$

The arrow runs one way only. Observing $R_1 > R_0$ does *not* let you conclude the price went up — quantity might have risen instead.

Example

An equivalence. Let revenue and cost be

$$R = pq, \quad C(q) = C_f + C_v q,$$

with C_f the fixed cost and C_v the variable cost per unit. A company reaches its break-even point exactly when revenue equals total cost:

$$\text{break-even reached} \iff pq = C_f + C_v q.$$

Here the arrow runs both ways: break-even implies the equation, and the equation implies break-even. That is what $\iff$ asserts, and it is a strictly stronger claim than $\Rightarrow$.

3.2 Describing a set

A set can be defined in three main ways.

i Definition: the three descriptions

By enumeration — list the elements between braces:

$$A = \{1, 2, 3, 4\}.$$

The set A contains exactly the elements 1, 2, 3 and 4.

By a property (set-builder notation) — state the universe the elements come from, a vertical bar read “such that”, and the condition:

$$B = \{x \in \mathbb{R} \mid x \geq 0\}, \quad C = \{x \in \mathbb{R} \mid a \leq x \leq b\}.$$

B is the set of non-negative real numbers; C is the set of reals between a and b inclusive.

By an interval — for sets of consecutive real numbers only.

In words: name the members one by one, state the rule for membership, or — if the members form an unbroken stretch of the number line — give the two endpoints.

Enumeration is only practical for small finite sets. Set-builder notation is the workhorse, and it is worth being fussy about its three parts, because omitting the universe is the most common error in the topic:

$$\underbrace{\{x \in \mathbb{R} \mid}_{\text{where } x \text{ lives}} \quad \underbrace{x \geq 0}_{\text{the condition}} \}$$

Write $\{x \mid x \geq 0\}$ without the $\in \mathbb{R}$ and you have not said whether you mean the non-negative integers, the non-negative reals, or something else. The universe is not decoration.

Key point. Set-builder notation matters because it is how you state precisely which values a variable or a decision is allowed to take — exactly what the constraint set of the production model did.

3.2.1 Intervals

Notation	Set-builder form	Endpoints included?
$[a, b]$	$\{x \in \mathbb{R} \mid a \leq x \leq b\}$	both
(a, b)	$\{x \in \mathbb{R} \mid a < x < b\}$	neither
$[a, b)$	$\{x \in \mathbb{R} \mid a \leq x < b\}$	left only
$(a, b]$	$\{x \in \mathbb{R} \mid a < x \leq b\}$	right only
$[a, +\infty)$	$\{x \in \mathbb{R} \mid x \geq a\}$	left only
$(-\infty, a]$	$\{x \in \mathbb{R} \mid x \leq a\}$	right only

The pattern: **a square bracket admits the endpoint, a round bracket excludes it**. Infinity always takes a round bracket, because $+\infty$ is not a number and cannot be an element of anything.

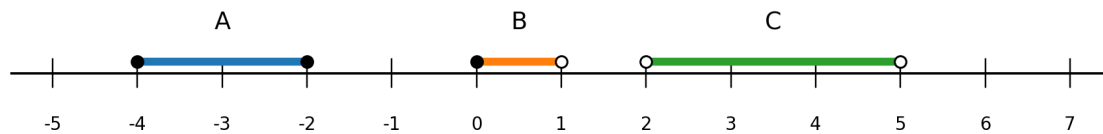


Figure 3.1: Intervals represented on the real line

In Figure 3.1, $A = [-4, -2]$, $B = [0, 1)$ and $C = (2, 5)$. The number -3 is an **interior point** of A : it lies in A with a little room on both sides, unlike the endpoints -4 and -2 . Note that B contains 0 but not 1, and that C contains neither 2 nor 5 — the shading conventions in the figure (filled dot included, hollow dot excluded) say the same thing the brackets do.

⚠ Common pitfall

You may see the open interval written $]a, b[$, with the brackets turned outwards. This is the standard French convention and means exactly the same thing as (a, b) . Some course slides use it. This course, and essentially all statistical software, uses (a, b) .

3.3 The standard number sets

Notation	Name	Example of use
$\mathbb{N}$	natural numbers	$0, 1, 2, 3, \dots$; number of customers, products, periods
$\mathbb{Z}$	integers	$\dots, -2, -1, 0, 1, 2, \dots$; integer changes up or down
$\mathbb{Q}$	rational numbers	fractions p/q with $p, q \in \mathbb{Z}$, $q \neq 0$; exact proportions
$\mathbb{R}$	real numbers	numbers on the real line; return, temperature, profit

Notation	Name	Example of use
$\mathbb{R}_+$	non-negative reals	$[0, +\infty)$; prices, costs, revenues, probabilities
$\mathbb{R}^p$	real vectors of dimension p	$\mathbf{x} = (x_1, \dots, x_p)$; one observation on p variables
$\mathbb{R}^{n \times p}$	real $n \times p$ matrices	a whole data set: n observations, p variables

These nest: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Every integer is rational (write -3 as $-3/1$) and every rational is real. The reverse fails: $\sqrt{2}$ and π are real but not rational.

Common pitfall

$0 \in \mathbb{N}$ in this course. Many textbooks start the naturals at 1. Ours starts at 0, so “0 is a natural number” is true here. This is a convention, not a fact about the world, which is exactly why the course fixes one and states it.

Relatedly, $\mathbb{R}_+ = [0, +\infty)$ **includes** zero. For strictly positive reals write $(0, +\infty)$ and say so.

3.4 Absolute value of a number

Definition: absolute value

The absolute value of a real number a , written $|a|$, is

$$|a| = \begin{cases} a, & \text{if } a \geq 0, \\ -a, & \text{if } a < 0. \end{cases}$$

In words: strip the sign. The absolute value is how far the number is from zero, and distance is never negative.

For example,

$$|13| = 13, \quad |-13| = -(-13) = 13, \quad \left|-\frac{1}{2}\right| = \frac{1}{2}, \quad |0| = 0.$$

Note that $-a$ is positive when a is negative, so the minus sign in the second case of the definition is not a typo.

3.4.1 Distance between two real numbers

! Key result: distance and tolerance

Let x_1 and x_2 be real. The distance between them on the real line is

$$d(x_1, x_2) = |x_1 - x_2|.$$

Distance is always non-negative, and it is zero exactly when the two numbers coincide:

$$d(x_1, x_2) \geq 0, \quad d(x_1, x_2) = 0 \iff x_1 = x_2.$$

For $a > 0$,

$$|x| < a \iff -a < x < a \iff x \in (-a, a), \quad |x| \leq a \iff x \in [-a, a].$$

In words: “ x is within a of zero” and “ x lies in the interval of width $2a$ centred at zero” say the same thing. Absolute-value inequalities are interval statements in disguise.

Shifting the centre gives the form you will use most: $|u - v| \leq c$ means $u \in [v - c, v + c]$. This single identity converts a large class of tolerance requirements — a machined part, a risk score, a revenue forecast — into intervals you can compute with.

3.5 Summation notation (sigma)

Summation notation is shorthand for addition. It earns its place because it lets you write a sum whose length is a symbol rather than a number, and because it makes the algebra of sums visible.

i Definition: the summation symbol

$$\sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n.$$

In words: “add up the a_i , letting i run from 1 to n .” The letter i is the **index**, the numbers below and above $\sum$ are the **limits**, and a_i is the **summand**.

Economists frequently use census data. Suppose a country is divided into six regions and N_i denotes the population of region i . The country’s total population is then

$$N_1 + N_2 + N_3 + N_4 + N_5 + N_6 = \sum_{i=1}^6 N_i,$$

read as “the sum of the N_i values, for i ranging from 1 to 6”.

The index is a placeholder with no meaning outside the sum: $\sum_{j=1}^n a_j$ and $\sum_{k=1}^n a_k$ denote the same number. This matters more than it looks — an index that also appears *outside* its sum is a symptom of an error.

Reading in both directions is the skill to build:

$$\sum_{i=1}^4 x_i = x_1 + x_2 + x_3 + x_4, \quad \sum_{j=2}^5 (2j-1) = 3 + 5 + 7 + 9 = 24, \quad \sum_{k=1}^n c = \underbrace{c + \cdots + c}_{n \text{ terms}} = nc.$$

That last one is worth staring at. The summand c does not mention k , so nothing changes as k advances; you simply add c to itself n times.

3.5.1 Example: price index

Suppose the prices of several goods vary over time. Price indices summarise the overall effect of these changes.

Example

Consider a basket containing n goods. For each good $i = 1, \dots, n$ let

q_i = quantity of good i in the basket, p_i^0 = unit price in the base year 0, p_i^t = unit price in year t .

Cost of the basket in the base year. Multiply each price by its quantity and add:

$$p_1^0 q_1 + p_2^0 q_2 + \cdots + p_n^0 q_n = \sum_{i=1}^n p_i^0 q_i.$$

Cost of the same basket in year t . The quantities are held fixed — that is the whole point of an index, to isolate price movement from changes in what people buy:

$$\sum_{i=1}^n p_i^t q_i.$$

The index. The base year is set equal to 100, so the index for year t is

$$I_t = \frac{\sum_{i=1}^n p_i^t q_i}{\sum_{i=1}^n p_i^0 q_i} \times 100.$$

Interpretation.

- If $I_t = 100$, the cost of the basket has not changed.
- If $I_t > 100$, it has increased.
- If $I_t < 100$, it has decreased.
- If $I_t = 150$, the basket costs 150% of its base-year cost — prices have risen by $150 - 100 = 50\%$.

The subtraction in that last line is the step people skip. An index of 150 is not a 150% increase; it is a 50% increase.

3.5.2 Summation rules

! Key result: algebra of sums

For any numbers a_i, b_i and any constant c (a value not depending on the index):

$$\textbf{Additivity: } \sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$$

$$\textbf{Homogeneity: } \sum_{i=1}^n c a_i = c \sum_{i=1}^n a_i$$

$$\textbf{Constant: } \sum_{i=1}^n c = nc$$

In words: sums split across addition, constants slide out front, and a constant summand simply gets multiplied by the number of terms.

The first two together are called **linearity**, and they are why summation notation is worth learning: they let you manipulate sums without ever expanding them.

! Common pitfall

There is no product rule and no rule for squares. In general

$$\sum a_i b_i \neq \left(\sum a_i \right) \left(\sum b_i \right), \quad \sum a_i^2 \neq \left(\sum a_i \right)^2.$$

A two-term counterexample settles it: with $a = (1, 2)$, $\sum a_i^2 = 1 + 4 = 5$ while $(\sum a_i)^2 = 3^2 = 9$. Linearity covers addition and constant multiples — nothing else.

3.5.3 Example: arithmetic mean and deviations from the mean

i Definition: arithmetic mean

The arithmetic mean of n numbers $x_1, x_2, \dots, x_n$ is their sum divided by n :

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The quantity $x_i - \bar{x}$ is the **deviation** of observation i from the mean.

Multiplying through by n gives a form used constantly in derivations:

$$\sum_{i=1}^n x_i = n\bar{x}.$$

! Key result: deviations sum to zero

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

Derivation. Split the sum using additivity, then handle each piece:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x}.$$

The second sum has a constant summand — $\bar{x}$ is one fixed number, not something that changes with i — so by the constant rule it equals $n\bar{x}$:

$$= \sum_{i=1}^n x_i - n\bar{x}.$$

But $\sum x_i = n\bar{x}$, so this is

$$= n\bar{x} - n\bar{x} = 0. \quad \blacksquare$$

In words: the mean sits exactly at the balance point of the data. Whatever lies above it is matched, in total, by what lies below. This is why raw deviations are useless as a measure of spread — they always total zero however scattered the data — and hence why we square them.

! Key result: computational form of the sum of squares

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2.$$

Derivation. Expand the square inside the sum:

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2).$$

Split into three sums by additivity:

$$= \sum_{i=1}^n x_i^2 - \sum_{i=1}^n 2x_i\bar{x} + \sum_{i=1}^n \bar{x}^2.$$

In the middle sum $2\bar{x}$ is constant, so pull it out by homogeneity; in the last, $\bar{x}^2$ is constant, so apply the constant rule:

$$= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + n\bar{x}^2.$$

Substitute $\sum x_i = n\bar{x}$ into the middle term:

$$= \sum_{i=1}^n x_i^2 - 2\bar{x}(n\bar{x}) + n\bar{x}^2 = \sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2,$$

and combine the last two terms:

$$= \sum_{i=1}^n x_i^2 - n\bar{x}^2. \quad \blacksquare$$

In words: you can get the sum of squared deviations without computing a single deviation — square the raw values, add them, and subtract n times the squared mean.

⚠ Common pitfall

μ and $\bar{x}$ are not the same object. Some sources write μ_X for the mean of a list of numbers. In statistics μ conventionally denotes a *population* mean — a fixed, usually unknown quantity — while $\bar{x}$ denotes the mean *computed from an observed sample*. These notes use $\bar{x}$ throughout for data you can see, and reserve μ for the population. The distinction becomes essential the moment estimation appears.

3.5.4 Double sums

When data are arranged in a grid, two indices are needed: a_{ij} for row i , column j . Consider a rectangular array with m rows and n columns, and let S_i be the total of row i :

$$\begin{array}{ccccccc} a_{11} & a_{12} & \cdots & a_{1n} & \longrightarrow & \sum_{j=1}^n a_{1j} & = S_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & \longrightarrow & \sum_{j=1}^n a_{2j} & = S_2 \\ \vdots & \vdots & & \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} & \longrightarrow & \sum_{j=1}^n a_{mj} & = S_m \end{array}$$

Adding the row totals gives the grand total, and the same grand total is reached by adding column totals instead:

! Key result: order of summation

$$S_1 + S_2 + \cdots + S_m = \sum_{i=1}^m S_i = \sum_{i=1}^m \sum_{j=1}^n a_{ij} = \sum_{j=1}^n \sum_{i=1}^m a_{ij}.$$

In words: totalling by rows and totalling by columns give the same answer — you are adding the same numbers in a different order.

⚠ Common pitfall

Reusing an index. In $\sum_i \sum_i a_{ii}$ the two sums collide and the expression is meaningless; each sum needs its own letter. Likewise, if the index survives into your final answer — “ $\sum_{i=1}^n x_i = nx_i$ ” — something has gone wrong.

3.6 Subsets, cardinality, and set operations

i Definition: subset

A set A is a **subset** of B if every element of A also belongs to B :

$$A \subseteq B \iff \forall x, x \in A \Rightarrow x \in B.$$

In words: A sits entirely inside B . Nothing in A escapes.

Two further terms:

- The **universal set** U is the reference set within which we are working.
- The **cardinality** of a finite set A , written $|A|$, is its number of elements.

So $|\{1, 2, 3, 4\}| = 4$, and $|\emptyset| = 0$ where $\emptyset$ is the **empty set**.

Let A and B be two subsets of a universal set U .

Operation	Notation	Definition	Management interpretation
Union	$A \cup B$	$\{x \in U \mid x \in A \text{ or } x \in B\}$	customers in at least one of the two segments
Intersection	$A \cap B$	$\{x \in U \mid x \in A \text{ and } x \in B\}$	customers in both segments simultaneously
Difference	$A - B = A \cap B^c$	$\{x \in U \mid x \in A \text{ and } x \notin B\}$	customers in A but not in B
Complement	$A^c = U - A$	$\{x \in U \mid x \notin A\}$	customers in the universe who are not in A

The “or” in the union is **inclusive**: x may be in A , in B , or in both. Everyday English often uses “or” exclusively (“tea or coffee” usually rules out both), which is a real source of confusion. Mathematical “or” never does.

! Key result: the size of a union (inclusion–exclusion)

For finite sets A and B ,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

In words: add the two sizes, then subtract the overlap once — anything in both sets got counted twice when you added.

! Common pitfall

Adding segment sizes without subtracting the overlap. “Segment A has 600 customers and segment B has 500, so we can reach 1,100” is wrong whenever the segments overlap. If 300 belong to both, the reachable total is $600 + 500 - 300 = 800$. This error is endemic in campaign reporting, where it inflates reach. The two counts add cleanly only when $A \cap B = \emptyset$.

3.6.1 Cartesian product and multidimensional observations

i Definition: Cartesian product

The Cartesian product of A and B is the set of ordered pairs whose first component belongs to A and whose second belongs to B :

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

In words: every possible pairing of something from A with something from B . Order matters — (a, b) and (b, a) are different pairs.

Repeating this construction gives $\mathbb{R}^p$. In a data set an individual is usually described by several variables at once, and is therefore represented by a vector:

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip}) \in \mathbb{R}^p,$$

where p is the number of variables measured for individual i .

Example

A Netflix user may be represented by a vector containing age, the number of hours watched per week, a preferred movie genre encoded numerically, the number of series abandoned before completion, and the user’s average rating of content. This five-dimensional vector, with $p = 5$ variables, lets a recommendation algorithm compare the user with others who have similar profiles.

Note what the encoding step is doing: “preferred genre” is not a number, so a numeric code is *imposed* on it. That is a modelling assumption, and if the codes are arbitrary — comedy = 1, drama = 2, horror = 3 — then any algorithm computing distances between users is quietly asserting that comedy is closer to drama than to horror.

Stacking n such observations gives the data set as a matrix

$$X = (x_{ij}) \in \mathbb{R}^{n \times p},$$

where x_{ij} is the value of variable j for observation i : rows are observations, columns are variables.

⚠ Common pitfall

Reading n and p off the wrong axis. A data set in $\mathbb{R}^{100 \times 8}$ has 100 observations and 8 variables, not the other way round. In these notes n *always* means the number of observations and p *always* the number of variables; no other letter is used for either.

3.6.2 Partition of a universal set

i Definition: partition

Let $X_1, X_2, \dots, X_k$ be non-empty subsets of U . The family $\{X_i\}_{i \in \{1, 2, \dots, k\}}$ forms a **partition** of U if

$$\bigcup_{i=1}^k X_i = U \quad \text{and} \quad X_i \cap X_j = \emptyset \quad \text{for every } i \neq j,$$

where

$$\bigcup_{i=1}^k X_i = X_1 \cup X_2 \cup X_3 \cup \dots \cup X_k.$$

In words: the subsets carve U into pieces with no gaps and no overlaps — every element of U lands in exactly one piece.

The phrase “exactly one” is the whole content of the definition. The union condition gives *at least one*; the disjointness condition gives *at most one*. Together: exactly one.

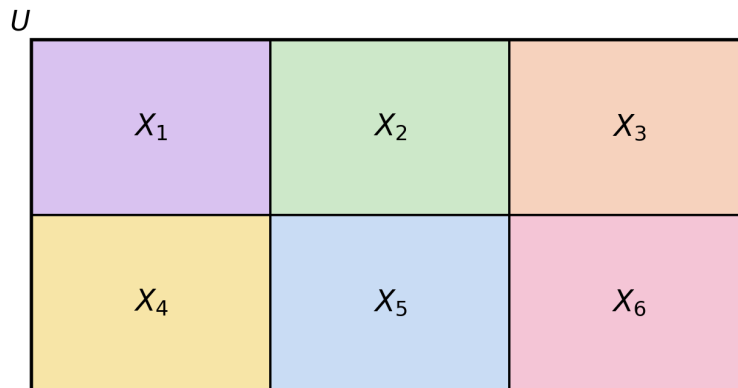


Figure 3.2: Partition of a universal set into six non-empty subsets

Partitions are what a categorical variable does to a data set. Assigning every customer a tier, every transaction a region, or every ticket a priority partitions the population — provided the categories are built correctly.

 Common pitfall

Boundary values landing in two classes. Suppose customers are banded by annual spend into $[0, 500]$, $[500, 2000]$ and $[2000, \infty)$, with square brackets throughout. A customer spending exactly \$500 qualifies for the first *and* the second, and one spending \$2000 for the second *and* the third. That is not a partition: the bands sum to more than the customer base, and a customer's band depends on which rule the query happens to evaluate first.

The fix is to close on the left and open on the right — $[0, 500)$, $[500, 2000)$, $[2000, \infty)$ — for every band except the last. This is the convention nearly all statistical software follows when binning a continuous variable, and it is worth adopting as a habit.

3.6.3 The simplex

One particular set recurs often enough — in weights, proportions, market shares, predicted class probabilities — to deserve a name.

 Definition: the simplex

The **simplex** in $\mathbb{R}^p$ is

$$\Delta_p = \left\{ \mathbf{w} \in \mathbb{R}_+^p \mid \sum_{j=1}^p w_j = 1 \right\}.$$

In words: the set of vectors with p non-negative components adding to exactly 1 — that is, all possible ways of splitting a whole into p shares.

Both conditions are needed. Non-negativity alone permits shares summing to 3; summing to 1 alone permits a negative share. The subscript p counts *components*.

 Common pitfall

Superscript versus subscript on Δ . You will meet Δ^2 , Δ_3 and Δ^{K-1} in different sources, all describing three shares summing to one. The disagreement is over whether the index counts components (3) or dimensions (2 — a triangle is two-dimensional even though it sits in three-dimensional space). This course always uses the subscript and always counts **components**: Δ_p has p of them.

4 Essential algebra for working with models

Algebra is what lets you transform the relationships above: simplify an expression, isolate a variable, solve an equation, linearize a relationship, or prepare an optimization problem. The operations are ones you have seen; what is worth rebuilding is the fluency to apply them without hesitation, because in later weeks they will be steps inside a larger argument rather than the point of the exercise.

4.1 Algebraic expressions

An algebraic expression combines numbers, variables and operations. The basic manipulations:

Operation	Formula	Purpose
Combine like terms	$2x + 3x = 5x$	group like terms
Expand	$a(b + c) = ab + ac$	remove parentheses
Factor	$ab + ac = a(b + c)$	factor out a common factor
Square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$	simplify quadratic functions
Difference of squares	$a^2 - b^2 = (a - b)(a + b)$	solve or factor expressions

Common pitfall

$(a + b)^2 \neq a^2 + b^2$. The cross term $2ab$ is not optional — test with $a = b = 1$: the left side is 4, and $a^2 + b^2 = 2$. The same error in other costumes: $\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$, and $\frac{1}{a + b} \neq \frac{1}{a} + \frac{1}{b}$.

4.2 Polynomials and quadratic equations

A polynomial of degree n is an expression of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0.$$

The condition $a_n \neq 0$ is what makes the degree well defined — it is the highest power that actually appears.

Examples.

- $P(x) = 3x + 1$: a polynomial of degree 1.
- $P(x) = 2x^2 - 3x + 1$: a polynomial of degree 2.
- $Q(x) = 5x^3 + x - 7$: a polynomial of degree 3. Note that the x^2 term is simply absent, which is allowed; only the *leading* coefficient must be non-zero.

4.2.1 Roots of a first-degree polynomial

A linear equation in one unknown is written

$$ax + b = 0, \quad a \neq 0,$$

with the single solution

$$x = -\frac{b}{a} = -a^{-1}b.$$

The condition $a \neq 0$ matters. If $a = 0$ the equation reads $b = 0$, which is either always true (when $b = 0$, so every x works) or never true (when $b \neq 0$, so no x does). Dividing by a without checking it is non-zero is the standard way to lose a case.

4.2.2 Roots of a quadratic polynomial

i Definition: quadratic equation and discriminant

A quadratic equation is written

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

and its **discriminant** is

$$\Delta = b^2 - 4ac.$$

! Notation clash — read once

The letter Δ is doing two unrelated jobs in this course. Bare Δ , as here, is the **discriminant** of a quadratic — a single number. Subscripted Δ_p , from the previous section, is the **simplex** — a set of vectors. They have nothing to do with each other; the alphabet is simply short. The subscript is what tells them apart, so read it.

! Key result: solutions of a quadratic

- If $\Delta > 0$, there are two distinct real roots:

$$x_1 = \frac{-b - \sqrt{\Delta}}{2a}, \quad x_2 = \frac{-b + \sqrt{\Delta}}{2a}.$$

- If $\Delta = 0$, there is one repeated real root:

$$x = -\frac{b}{2a}.$$

- If $\Delta < 0$, there is no real root.

In words: the discriminant is a preview. Compute it first and you know how many answers to expect before doing any work — and whether a real answer exists at all.

The repeated root $-b/(2a)$ is also the **vertex** of the parabola $y = ax^2 + bx + c$ in every case, sitting midway between the two roots when they exist. When $a < 0$ the parabola opens downward and the vertex is a maximum, which is why quadratics appear in profit problems.

4.3 Exponents and logarithms

For $a, b > 0$ and compatible exponents, the basic rules are

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn},$$

$$(ab)^n = a^n b^n, \quad a^{-n} = \frac{1}{a^n}, \quad a^0 = 1.$$

For equations with the same positive base $a \neq 1$,

$$a^f = a^g \iff f = g, \quad \text{and in particular} \quad e^f = e^g \iff f = g.$$

Common pitfall

$a^m a^n = a^{m+n}$, **not** a^{mn} . Exponents add when you multiply powers, and multiply only when you raise a power to a power. Check with small numbers whenever unsure: $2^2 \cdot 2^3 = 4 \cdot 8 = 32 = 2^5$, and $5 = 2 + 3$.

Definition: logarithm

The logarithm to base $a > 0$, with $a \neq 1$, is defined by

$$\log_a(x) = y \iff a^y = x.$$

The **natural logarithm** is written $\ln$ and corresponds to base e :

$$\ln(x) = y \iff e^y = x.$$

In words: a logarithm answers “what power do I raise the base to, in order to get x ?” It is the exponential run backwards.

Useful rules, all of which follow from the exponent rules because logarithms *are* exponents:

$$\ln(xy) = \ln x + \ln y, \quad \ln\left(\frac{x}{y}\right) = \ln x - \ln y, \quad \ln(x^r) = r \ln x,$$

$$\ln(e) = 1, \quad \ln(1) = 0, \quad \ln(e^x) = x, \quad e^{\ln x} = x \quad (x > 0).$$

And for equations,

$$\log_a(f) = \log_a(g) \iff f = g, \quad \text{in particular} \quad \ln(f) = \ln(g) \iff f = g.$$

The rule $\ln(x^r) = r \ln x$ is the one that does the work in this course: it is how an unknown stuck in an exponent gets brought down where you can solve for it.

⚠ Common pitfall

Two distinct traps.

The notation. “log” is ambiguous. In R and most mathematical writing, $\log(x)$ is the natural logarithm; in many spreadsheets, calculators and engineering texts, **LOG** means base 10. These notes always write $\ln$ for the natural log and always show the base otherwise. When reading anything else, check.

The algebra. There is no rule for the logarithm of a sum:

$$\ln(x + y) \neq \ln x + \ln y.$$

The left side does not simplify at all. What $\ln x + \ln y$ equals is $\ln(xy)$ — a product, not a sum. This is probably the most common algebra error in the course.

4.3.1 Applied example: exponential growth and decay

! Key result: growth and decay factors

An initial quantity Q_0 that **increases** by $p\%$ each year has, after t years, the value

$$Q_0 \left(1 + \frac{p}{100}\right)^t.$$

An initial quantity Q_0 that **decreases** by $p\%$ each year has, after t years, the value

$$Q_0 \left(1 - \frac{p}{100}\right)^t.$$

The number $1 + p/100$ is the **growth factor** associated with an increase of $p\%$; the number $1 - p/100$ is the **decay factor** associated with a decrease of $p\%$. Here Q_0 is the initial value, p the annual rate as a percentage, and t the number of years.

The factor is *multiplied* repeatedly rather than added because each year’s change applies to the new total, not to the original. That is what “compound” means, and it is the entire content of the formula.

Example

A depreciating asset. A new car was purchased for \$30,000. Assume its value decreases by 15% per year. What will its value be after six years?

The rate is a decrease, so the decay factor is

$$1 - \frac{15}{100} = 0.85.$$

After six years the value is

$$30000 \times (0.85)^6 = 30000 \times 0.377150 \approx \$11,314.49.$$

A common wrong answer is $30000 \times (1 - 6 \times 0.15) = 30000 \times 0.10 = \$3,000$, which subtracts 15% of the *original* value six times. Depreciation compounds: each year removes 15% of whatever is left. Stepping through confirms it: $\$30,000 \rightarrow \$25,500 \rightarrow \$21,675 \rightarrow \$18,423.75 \rightarrow \$15,660.19 \rightarrow \$13,311.16 \rightarrow \$11,314.49$.

Example

Solving for the rate. A sum of \$5,000 deposited in an account grew to \$10,000 after 15 years. What constant annual interest rate p was applied?

Set up the growth equation with the rate unknown:

$$5000 \left(1 + \frac{p}{100}\right)^{15} = 10000.$$

Divide both sides by 5,000. Note that the initial deposit cancels — the doubling time does not depend on how much you started with:

$$\left(1 + \frac{p}{100}\right)^{15} = 2.$$

The unknown sits in the base with a known exponent, so raise both sides to the power $1/15$:

$$1 + \frac{p}{100} = 2^{1/15}.$$

Equivalently, and more usefully when the exponent is the unknown, take natural logarithms first:

$$15 \ln\left(1 + \frac{p}{100}\right) = \ln 2 \quad \Rightarrow \quad \ln\left(1 + \frac{p}{100}\right) = \frac{\ln 2}{15} = \frac{0.6931}{15} \approx 0.046210.$$

Exponentiating both sides,

$$1 + \frac{p}{100} = e^{0.046210} \approx 1.047294,$$

so

$$\frac{p}{100} \approx 0.047294 \quad \Rightarrow \quad p \approx 4.73\%.$$

Check. $5000 \times (1.047294)^{15} = 5000 \times 2.0000 = \$10,000$. Correct.

5 Proof techniques, propositions, and quantifiers

i Developed in class

This section is built during the lecture rather than before it. It covers propositions and quantifiers ($\forall$, $\exists$, $\exists!$), implication and equivalence, necessary and sufficient conditions, and the five proof strategies: direct proof, proof by contrapositive, proof by contradiction, mathematical induction, and disproof by counterexample.

Notes will be added here after the session.

One thing is worth flagging in advance, because it trips up anyone who has written code. The symbol $\exists!$ means “**there exists exactly one**” — the $!$ marks *uniqueness*, not negation. It is a stronger claim than $\exists$, not its opposite:

$\exists x P(x)$ means at least one x works;

$\exists! x P(x)$ means exactly one x works.

For instance $\exists x \in \mathbb{R}$ with $x^2 = 4$ is true — both 2 and -2 qualify — but $\exists! x \in \mathbb{R}$ with $x^2 = 4$ is **false**, precisely because there are two. Negation is written $\neg$, or $\nexists$ for “there does not exist”.

One-minute recap

🔥 If you remember nothing else

- **Variable or parameter?** Ask whether you know the number before you start solving. Known in advance $\Rightarrow$ parameter. The same symbol can switch roles when the question changes.
- **When a resource is scarce, rank options by return per unit of that resource**, not per unit of output. Product A wins on profit per unit and loses on profit per labor hour.
- **Set-builder notation needs a universe.** $\{x \in \mathbb{R} \mid x \geq 0\}$, not $\{x \mid x \geq 0\}$.
- **Brackets:** square includes the endpoint, round excludes it. Infinity always gets round.
- $|u - v| \leq c$ is the interval $[v - c, v + c]$. Every tolerance statement is an interval statement.
- $|A \cup B| = |A| + |B| - |A \cap B|$ — subtract the overlap or you double-count.
- **A partition means exactly one.** Use half-open bands $[a, b)$ so boundary values land in one bin only.
- $\sum$ is **linear and nothing more:** it splits over $+$ and lets constants out. There is no product rule.
- $\sum(x_i - \bar{x}) = 0$ always. Use it as an arithmetic check on any mean you compute.
- **An index of 150 is a 50% increase**, not a 150% one.
- **Logs turn powers into products and products into sums** — that is how you get an unknown out of an exponent. But $\ln(x + y)$ does not simplify.
- **Growth compounds.** Losing 15% a year for six years is $\times(0.85)^6$, not $\times(1 - 6 \times 0.15)$.
- **In this course:** $0 \in \mathbb{N}$; n = observations, p = variables; vectors are bold; $\ln$ is the natural log.